

L. T. MINH TRINH

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EDUCATION

University College London

MEng in Mechanical Engineering, Minor in Intelligent Systems

London, United Kingdom

Sep. 2023 - exp. Jun. 2027

- Grade: First Class Honours (on track)
- Extracurricular: UCL Racing - Mars Rover team, Mechanical Engineering Society, AI Society.

The National Mathematics and Science College

A-Level

Coventry, United Kingdom

Sep. 2021 - Jul. 2023

- Grade: A*, A* and A (respectively in Maths, Further Maths and Physics)
- Awarded full academic scholarship 135% tuition for consecutive years of A-Level.

EXPERIENCE

Autonomous and Control Engineer

UCL Racing - Mars Rover team

London, United Kingdom

Oct. 2023 - Present

- Leading transition of rover vision system from 2D to 3D SLAM, enhancing localisation accuracy and mapping capability for robust autonomous navigation.
- Developed computer vision system for Mars rover competing in European Rover Challenge (ERC), fine-tuning EfficientNet for obstacle/terrain classification (85% accuracy) and implementing ArUco marker detection for autonomous navigation and collision avoidance.
- Built embedded control system bridging high-level autonomy with rover hardware, programming Arduino microcontrollers in C++ to handle motor drivers, sensor interfaces (I2C/SPI), and ROS2 communication protocols for real-time mobility control.

Research Intern

FPT Software

Hanoi, Vietnam

Jun. 2024 - Aug. 2024

- Designed and simulated 6-DOF robotic arm for automated sorting system, implementing analytical IK that achieved 100% solve rate within joint limits, enabling reliable multi-object sorting operations.
- Integrated pre-trained ResNet50 with depth camera pipeline in ROS2-Isaac Sim framework, extracting 3D coordinates from RGB-D data to enable vision-guided pick-and-place for automated sorting tasks.

PROJECTS

Summer Research (NeurIPS Workshop 2025 - Poster)

Metacognitive Sensitivity for Test-Time: Dynamic Model Selection

London, United Kingdom

Jun. 2025 - Aug. 2025

- Developed metacognitive framework for test-time model selection using contextual bandits (UCB/TS), improving SOTA image classifier accuracy by 10% and vision-language models (CLIP/ALIGN) by 2% without any fine-tuning.
- Accelerated Bayesian inference from CPU-bound PyMC to GPU execution using JAX/Pyro, achieving 3x speed-up for demonstrations and 5x for full-scale experiments, enabling real-time model selection.
- First-author poster (**accepted**) at NeurIPS 2025 Workshop: confidence-based metacognitive signals enable effective test-time adaptation across diverse architectures without gradient updates.

TECHNICAL SKILLS

Languages

Python, C/C++, SQL, MATLAB

Frameworks & Libraries

PyTorch, TensorFlow, JAX, Scikit-learn, NumPy/Pandas

Tools

Docker, Git, Linux, ROS, OpenCV, Fusion 360, Isaac Sim/Lab